The Bacterial DNA-Directed RNA Polymerase Core Enzyme: A Mechanistic Review

Commissioned review synthesis. Scope: the species-neutral bacterial core enzyme ($\alpha_2\beta\beta'\omega$) that carries out DNA-templated RNA synthesis. Sigma factors, elongation factors, and promoter-specific regulators are explicitly out of scope.

1. Executive Summary

The bacterial DNA-directed RNA polymerase (RNAP) **core enzyme** is a five-polypeptide, four-species assembly — two identical α subunits, one β, one β′, and one ω — with the canonical composition **$\alpha_2\beta\beta'\omega$** (~380 kDa). It is the minimal machine that catalyzes template-directed phosphodiester bond formation; it is *sufficient* for RNA chain elongation but *not* for promoter-specific initiation, which requires a dissociable σ factor to form the holoenzyme (Eσ). The core is organized as a "crab-claw," with the two largest subunits (β and β′) forming the pincers and the catalytic cleft, a single catalytic Mg^{2+} held by an absolutely conserved β′ aspartate motif, the α dimer acting as the assembly platform on the back of the enzyme, and the small ω subunit clamping the β′ C-terminus to promote folding and stability (Zhang et al. 1999, P 10499798; Weiss et al. 2017, P 27799328).

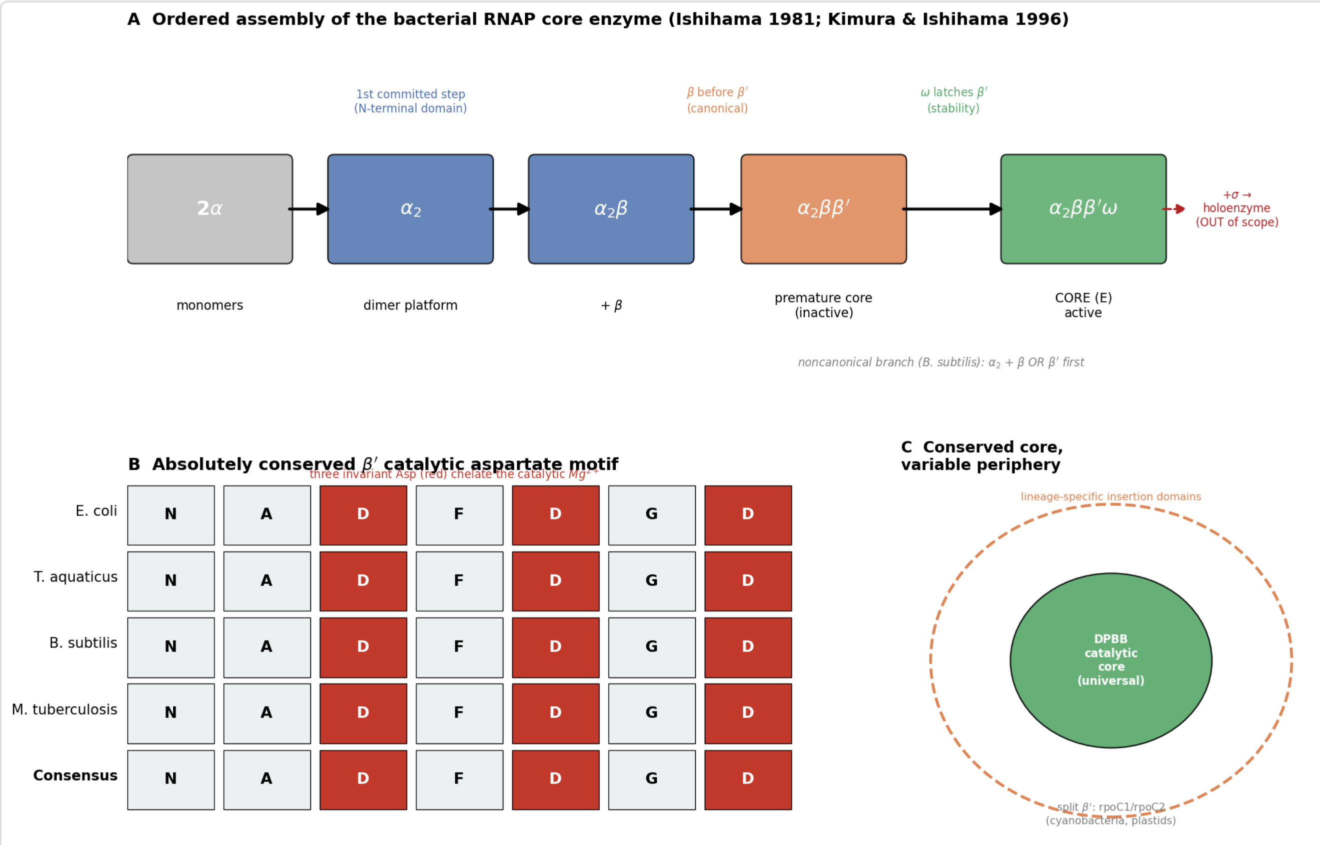

Figure 1. *Figure 1. (A) Ordered assembly pathway of the core enzyme, from a monomers through the α_2 platform and the premature core to the active $\alpha_2\beta\beta'\omega$ enzyme; the σ-bound holoenzyme (dashed) is downstream and out of scope. (B) The absolutely conserved β′ active-site aspartate motif (NADFDGD): the three invariant Asp residues chelate the single catalytic Mg^{2+}. (C) The "conserved core, variable periphery" principle — a universal double-psi-β-barrel (DPBB) catalytic core decorated by lineage-specific insertion domains, with genetic splitting of β′ (rpoC1/rpoC2) in cyanobacteria and plastids. Panel B is illustrative of the invariance emphasized by Zhang et al. 1999 (P 10499798).*

Assembly is an **ordered pathway**: $2\alpha \rightarrow \alpha_2 \rightarrow \alpha_2\beta \rightarrow \alpha_2\beta\beta' \rightarrow (\omega$, maturation$) \rightarrow$ active core E (Ishihama 1981, P 7015808; Kimura & Ishihama 1996, P 9078382). The catalytic core (the double-psi β-barrel active site) and the α-dimerization module are among the most deeply conserved protein features in biology, traceable toward the last universal common ancestor (LUCA) and homologous to the cores of archaeal and eukaryotic RNAPs (Fouqueau et al. 2017,

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P 28657884

Onuoha et al. 2026,

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P 42417148

Below we define the system's boundaries, lay out the best current mechanistic model, catalogue the molecular players, and mark the places where evidence is lineage-specific or genuinely unresolved.

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2. Definition and Biological Boundaries

2.1 What is included

The **core enzyme** is defined by composition and catalytic capability: - **α_2** — a homodimer; assembly scaffold and platform for regulatory contacts (via the mobile C-terminal domain, αCTD). - **β (rpoB)** and **β′ (rpoC)** — the two large subunits that build the DNA-binding channel, the catalytic center, the NTP-entry (secondary) channel, and the RNA-exit channel. - **ω (rpoZ)** — a small subunit wrapped around the β′ C-terminus; an assembly/stability factor.

The functional signature of the core is **processive, template-directed RNA synthesis** (nucleotide addition cycle), independent of promoter recognition.

2.2 What is deliberately excluded (and commonly confused)

- **σ factors** convert core → holoenzyme and confer promoter specificity. They are dissociable and are *not* part of the core; including them changes the biological question from "how is RNA made" to "where does transcription start." (Ishihama 1981 shows σ acts as a regulatory maturation/initiation factor, not a stoichiometric core subunit.)
- **Elongation/termination factors** — NusG/Spt5, NusA, GreA/B, Mfd, Rho — act *on* the core but are separate machines (e.g., Mfd loading dynamics: Brewer et al. 2026,

  ;

  P 42444603

  Rho spatial biology: Bossi et al. 2026,

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  P 42366572

  Spt5/NusG is the *only* transcription factor universally conserved across all three domains, and is still not a core subunit (Werner 2012, 22306403).

- **Second-messenger regulators** — (p)ppGpp/DksA bind the *assembled* enzyme (β′–ω interface, site 1; secondary channel, site 2) to reprogram transcription; they are regulatory ligands, not structural components (Ortiz-Vasco et al. 2024, 38574051).
- **Archaeal/eukaryotic nuclear RNAPs and organellar/single-subunit (phage-type) RNAPs** are evolutionarily related (multisubunit) or unrelated (T7-type) machines and should be treated separately, even though the multisubunit catalytic core is homologous.

2.3 Competing definitions

"Core enzyme" is used consistently to mean $\alpha_2\beta\beta'\omega$. Two definitional wrinkles exist: (i) historically, ω was sometimes regarded as a copurifying contaminant / non-essential subunit, so older "core" preparations were treated as $\alpha_2\beta\beta'$; modern usage includes ω. (ii) In some Gram-positive and specialized bacteria, additional small subunits (e.g., δ, ε) copurify with RNAP and are sometimes discussed alongside the "core," but they are lineage-specific accessory subunits, not part of the universal core module.

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3. Mechanistic Overview

3.1 Assembly: the obligatory ordered pathway

Reconstitution from purified E. coli subunits established a strictly stepwise route (Ishihama 1981, 7015808):

> $2\alpha \rightarrow \boldsymbol{\alpha_2} \rightarrow \boldsymbol{\alpha_2\beta} \rightarrow \boldsymbol{\alpha_2\beta\beta'}$ (premature core) → **E** (active core)

Key mechanistic points: 1. **α dimerization is the first committed step** and nucleates everything else, mediated by the α N-terminal domain (α-NTD, ≤ residue ~235). The interface is a defined side-chain cluster (e.g., F35, I46) conserved into Thermus (Kannan et al. 2001,

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P 11266593

Miyake et al. 1998,

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P 9477962

2. **β loads before β′.** The α-NTD presents distinct surfaces for β and β′ recruitment (Kimura & Ishihama 1996,

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P 9078382

Long-range allosteric coupling within the assembling complex is real: an α-NTD mutation defective for β′ contact is suppressed by a distant β′ substitution (G333D) (Sujatha et al. 2001,

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P 11212907

3. **Maturation to the active core** requires a conformational activation step; at low temperature the "premature core" ($\alpha_2\beta\beta'$) accumulates in an inactive but near-native conformation and matures only in the continued presence of σ or DNA (Ishihama 1981). 4. **ω joins to stabilize β′.** Its recruitment latches the β′ C-terminus, promoting correct folding and preventing β′ degradation/dissociation (Weiss et al. 2017,

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P 27799328

Obligatory vs. conditional vs. accessory: - *Obligatory:* α dimer formation → recruitment of the two large subunits → catalytic-cleft closure around a single Mg^{2+}. - *Conditional / lineage-variable:* the strict β-before-β′ order. In Bacillus subtilis a **noncanonical branch** exists in which the α dimer can associate with *either* β *or* β′ before completing the core (Tewary et al. 2026,

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P 42001948

- *Accessory (important but not catalytically obligatory):* ω incorporation; the core can assemble and function without ω, but is less stable and more degradation-prone.

3.2 Catalysis: the nucleotide addition cycle (NAC)

The catalytic center lies at the β/β′ interface on the back wall of the main channel, with a single Mg^{2+} (Mg-I) chelated by the absolutely conserved β′ **NADFDGD** aspartate triad (Zhang et al. 1999,

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P 10499798

Each cycle: 1. NTP delivery through the secondary channel to the insertion (i+1) site. 2. **Trigger-loop (TL) folding** closes over the correctly paired NTP, positioning it for catalysis; the

Rim-helices/F-loop couple to TL closure (Dhingra et al. 2026,

P 42378287

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3. Two-metal-ion phosphoryl transfer extends the RNA by one nucleotide; pyrophosphate release. 4. **Bridge-helix / clamp** conformational oscillations drive single-nucleotide translocation; TL unfolds, resetting the cycle.

The clamp (largely β′) can open and close by tens of Å — cryo-EM of E. coli core versus the Taq crystal structure revealed ~25 Å channel opening — and these motions are functionally required and universally conserved across bacterial, archaeal, and eukaryotic RNAPs (Darst et al. 2002,

P 11904365

; Pilotto & Werner 2022,

P 36144426

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4. Major Molecular Players and Active Assemblies

Subunit (gene)	Copies	Family / role	Key mechanistic contributions
α (rpoA)	2	α-subunit family; protein dimerization	α-NTD forms the α_2 assembly platform and nucleates core assembly; presents β- and β′-binding surfaces. The **αCTD** is a mobile, flexibly tethered domain that stimulates initiation by binding UP-element DNA and activators (Fis, SoxS, TyrR, CAP) and by contacting σ region 4.2 — a *regulatory* interface outside the core's catalytic remit (Kimura & Ishihama 1996; Kannan et al. 2001; Ross et al. 2003, P 12756230; Aiyar et al. 2002, P 11866514). The functional split between α-NTD (assembly) and αCTD (regulation) within one subunit is a clean illustration of where the core module's boundary lies.
β (rpoB)	1	β-subunit family (double-psi β-barrel)	Forms one pincer of the crab claw and part of the catalytic cleft, NTP-binding, and the rifampicin pocket adjacent to the active site. Rif-resistance mutations (e.g., βS450L in M. tuberculosis) cluster in conserved rpoB regions (Zhang et al. 1999; Eckartt et al. 2026, P 42156950).
β′ (rpoC)	1	β′-subunit family (double-psi β-barrel)	Second pincer; carries the catalytic Mg^{2+} (NADFDGD), the bridge helix, the trigger loop, and the mobile clamp; contains the σ-binding coiled-coil (β′ 260–309 in E. coli) (Zhang et al. 1999; Arthur et al. 2000, P 10764785).
ω (rpoZ)	1	ω-subunit family; RPB6 homolog	Wraps the β′ C-terminus; chaperones β′ folding and retains it in the assembled enzyme; provides part of the (p)ppGpp site-1 surface at the β′–ω interface (Weiss et al. 2017; Chatterji et al. 2007, P 17233676).

Active assemblies along the pathway: α (monomer) → α_2 (platform) → $\alpha_2\beta$ → $\alpha_2\beta\beta'$ (premature core; catalytically inert, near-native fold) → **$\alpha_2\beta\beta'\omega$ = core (E)** → +σ → **Eσ holoenzyme** (initiation-competent; out of scope but the immediate downstream product).

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5. Evolutionary and Cell-Biological Variation

5.1 Deep conservation and origin

The **catalytic core is LUCA-deep.** All cellular (multisubunit) RNAPs share a two-**double-psi β-barrel (DPBB)** active-site configuration and a homologous β/β′ catalytic module; this architecture is traceable across all three domains and is one of the strongest molecular arguments for a single origin of cellular transcription (Burton 2014, P 25764332; Fouqueau et al. 2017, P 28657884). Bacterial β and β′ are single polypeptides; in archaea and eukaryotes each is split into two (Rpo/Rpb) chains, but the fold and active-site geometry are preserved.

5.2 The α platform and its representatives

The **α dimerization module** is ancient. It began as a *homodimeric* scaffold in bacteria (best represented by bacterial α_2) and was later duplicated and partitioned into *asymmetric heterodimers* in archaea (Rpo3/Rpo11) and eukaryotes (RPB3/RPB11 for Pol II; paralogs for Pol I/III) (Onuoha et al. 2026, P 42417148). Thus the bacterial α homodimer is the best living representative of the ancestral assembly platform. ω is homologous to eukaryotic **RPB6** — an ancient, conserved small subunit — reinforcing that all four core species predate the bacterial/archaeal split.

5.3 Lineage- and state-specific variation

- **Assembly order is not universally linear:** the B. subtilis "α binds β *or* β′ first" branch shows the pathway topology varies between Gram-negative and Gram-positive lineages

(Tewary et al. 2026,

P 42001948
).

- **ω's regulatory role is lineage-variable:** ω contributes to (p)ppGpp binding and relA/stringent-response tuning in E. coli, but the key ppGpp-coordinating residues are *not* conserved in S. aureus, where rpoZ deletion does not impair starvation survival even though it destabilizes the enzyme (Weiss et al. 2017; Chatterji et al. 2007). Its structural chaperone role is conserved; its second-messenger role is not.
- **Subunit-architecture variation:** the *fold* is universal but its packaging is not. In cyanobacteria and chloroplast (plastid-encoded) RNAPs the β′ subunit is split across two genes (rpoC1/rpoC2), i.e. the single bacterial polypeptide is encoded as two chains that reconstitute the same crab-claw geometry — an alternative genetic route to the same structural outcome. Superimposed on the conserved core are lineage-specific insertion/"sequence-insertion" domains within β and β′ that vary greatly in size between phyla (large in E. coli, minimal in Thermus), decorating but not altering the catalytic center (consistent with the conserved-core/variable-periphery picture of Fouqueau et al. 2017; Onuoha et al. 2026). *Caveat:* these architectural variants are established from comparative genomics/structures across a modest set of taxa and are noted here rather than exhaustively sampled.
- **Accessory small subunits** (δ, ε in Firmicutes) modulate core behavior in some Gram-positives but are absent in E. coli — variation at the *periphery* of the core, not its catalytic heart. In several Gram-positive and actinobacterial lineages, ancillary factors (e.g., the δ subunit, HelD-type ATPases, RbpA/CarD) associate with the core to recycle or stabilize it, but these are dissociable accessory proteins, not universal core subunits, and are treated here as adjacent to the core module.
- **Physiological/spatial states:** the assembled core partitions with the nucleoid and forms clusters that reorganize between exponential and stationary phase (Bossi et al. 2026,

  P 42366572
)
  — a cell-biological, not compositional, variation.

6. Constraints, Dependencies, and Failure Modes

Ordering constraints (what must precede what): 1. α_2 must form before the large subunits can be recruited — no productive α-monomer→β complex is the committed route. 2. In the canonical (enterobacterial) pathway, β loads before β′; the premature core $\alpha_2\beta\beta'$ must undergo a conformational **maturation** step to become catalytically active. 3. ω acts on β′ *after* β′ is in the complex — it is a stabilizer of an existing subunit, not an initiator of assembly.

Mutually exclusive / conditional events: - Core assembly and promoter-specific initiation are temporally separated: σ binding (holoenzyme formation) is downstream of core maturation and is reversible (σ cycling). - (p)ppGpp/DksA regulation requires an *already assembled* enzyme; it cannot substitute for assembly.

Failure modes (evidence-backed): - **Loss of ω:** β′ misfolding/degradation, δ/σ imbalance, and general RNAP dissociation, with downstream stress-resistance and biofilm defects (S. aureus; Weiss et al. 2017). This is the clearest demonstration that ω's job is to keep β′ folded and the core intact. - **Defective α–large-subunit contacts:** ts α mutants (e.g., R45C defective for β binding; C131A defective for β′ contact) arrest assembly at defined intermediates; some are rescued by second-site β′ suppressors, revealing long-range allosteric coupling (Sujatha et al. 2001,

P 11212907

). - **Catalytic-cleft perturbation:** rifampicin binding in the β pocket sterically blocks nascent RNA extension; resistance requires β mutations that also carry collateral fitness costs via transcription attenuation (Eckartt et al. 2026,

P 42156950

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What the evidence rules out: the historical view of ω as a dispensable contaminant is refuted — it has a definite β′-chaperone function. The idea that assembly could begin from a β/β′ heterodimer without α is not supported by reconstitution; α_2 is the obligatory nucleus.

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7. Controversies and Open Questions

1. **How linear is assembly, really?** The classic enterobacterial β-before-β′ order is challenged by the B. subtilis noncanonical branch (Tewary et al. 2026). Whether parallel routes coexist

in a single organism, and how dedicated assembly chaperones (e.g., the RbpA/HelD-type factors, GroEL) bias flux, remains open.

2. **Is ω essential?** ω is dispensable for viability in E. coli and S. aureus under standard conditions but is required for full stability and stress fitness. The boundary between "accessory" and "required" is condition- and lineage-dependent — a genuine definitional and biological ambiguity.
3. **Structural vs. regulatory ω:** the chaperone function appears universal; the (p)ppGpp-binding/stringent role is not conserved (E. coli vs. S. aureus). Disentangling these two roles across taxa needs more comparative structural and genetic work.
4. **Organism-mixing caveat:** much mechanistic detail comes from three model systems — E. coli (genetics/assembly), Thermus (crystallography), and Mycobacterium (drug resistance). Extrapolating quantitative claims (rates, stabilities, residue-level roles) across these and to under-studied phyla risks conflating lineage-specific features with universal ones. Several "universal" statements rest on structural homology plus a handful of organisms rather than broad experimental sampling.
5. **Active-site dynamics:** the precise coupling of trigger-loop folding, bridge-helix bending, and translocation is still being refined with new inhibitor-trapped structures (Dhingra et al. 2026); the energetic model of the NAC is not fully settled.

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8. Key References

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Uncertainty note: Quantitative and residue-level claims derive predominantly from E. coli, Thermus, and Mycobacterium. Universality statements about assembly order, ω function, and active-site dynamics rest on structural homology plus a limited set of model organisms and should not be over-generalized to all bacterial phyla.